MINOR IN COMPUTER SCIENCE

Rationale

The CMS department offers a Computer Science minor for RHU students. It is designed for students who want to include basic proficiency in computer science to enhance or supplement their major program.

Program Objectives

The aims of a minor in Computer Science are:

- Provide RHU graduates with basic proficiency in computer science
- Support RHU graduates with essential computing skills desired/demanded by the market

Learning Outcomes

At the end of this minor, the student is expected to demonstrate:

- An ability to analyze a problem, identify and define the computing requirements appropriate to its solution
- An ability to use current techniques, skills, and tools necessary for computing practices
- An ability to develop programming skills using modern programming languages

Entrance Requirements

This minor is offered to all RHU students except those in CCE and Business IT Management.

Early in their major, interested RHU students must complete the appropriate form declaring their intent to minor in Computer Science while completing their regular major.

Computer Science Minor Policies

- Obtain a Minor Cumulative Grade Point Average of no less than 70%
- Although common courses between student major requirements and Computer Science minor requirements are counted to fulfill the minor requirements, the student must complete a minimum of 6 credit hours of coursework that is not counted toward the requirements for their major or any other minor.

Career Options

This minor allows its holders to pursue careers in various sectors, regardless of their major. Graduates of this minor can seek jobs in software engineering, web and mobile development, database/system administration, network engineering/administration, and IT consultancy.

Curriculum and Program

The Computer Science minor consists of six courses (18 credits), of which two are mandatory and four are electives, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

I. Mandatory Courses

The two mandatory courses (6 credits) are:

COSC 214	Introduction to Programming	3(2,2)
COSC 215	Advanced Programming and Data Structures	3(2,2)

II. Elective Courses

The elective courses will be selected in consultation with the Computer Science minor advisor based on the student's needs and background. Students may choose four elective courses (12 credits) from the following list.

COSC 231	Database Management Systems	3(3,0)
COSC 316	Design and Analysis of Algorithms	3(3,0)
COSC 333	Web Programming	3(3,0)
COSC 341	Software Engineering	3(3,0)
COSC 351	Logic Design	3(3,0)
COSC 434	Advanced Web Programming	3(3,0)
COSC 435	Mobile Application Development	3(2,2)
COSC 436	Game Programming	3(3,0)
COSC 438	Introduction to Software Testing	3(3,0)
COSC 451	Operating Systems	3(3,0)
COSC 480	Machine Learning and Data Mining	3(3,0)
COSC 481	Artificial Intelligence	3(3,0)
COSC 482	Data Science and Web Scraping	3(3,0)
Any other approved CS elective course		

MINOR IN DATA ANALYTICS

Rationale

"Information is the oil of the 21st century, and analytics is the combustion engine" - P. Sondergaard.

Are you a student in Business, Engineering, or Science? Do you want to add a new dimension to your areas of expertise to boost your career? You can do more by enrolling in the new DATA ANALYTICS (DA) minor. The demand for data analytics expertise has grown rapidly over the past few years and is expected to grow even faster in the coming years. Students graduating with a data analytics minor are well positioned to bridge the gap in their chosen field.

Program's Purpose

Graduates of this minor will be able to use their acquired skills across various industries and in the non-profit and government sectors. Telecommunications, banking, financial services, the environmental sector, energy, biomedical, police, biology, bioinformatics, and physics are just a few areas in which you can operate. The minor shall also serve as good preparation for further and more advanced graduate and professional studies in Data Analytics and related areas.

Learning Outcomes

At the end of this minor, the student is expected to demonstrate the ability to:

- Use current techniques, architectures, and tools to collect and store data;
- Apply data analytics tools and techniques to support and improve decision-making;
- Design systems and tools to interpret and visualize data.

Entrance Requirements

This minor is offered to all RHU students except CCE students.

Early in their major, interested RHU students must complete the appropriate form declaring their intent to minor in Data Analytics while completing their regular major.

Program Requirements

This minor is structured to accommodate undergraduate RHU students in the BE or BS program from different disciplines (engineering, computer science, business). It may also

be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements.

Early in their studies, interested students must declare their intent to pursue a minor in DA by completing the relevant minor declaration form and informing their advisor.

To complete the DA minor, a student must

1) Declare a minor in Data Analytics by completing the Minor Declaration Form;
2) Obtain the approval of the dean of the student's college major and the dean of the College of Arts and Sciences;
3) Earn a Minor Cumulative Grade Point Average of no less than 70%;
4) Complete 18 credits of coursework as specified below.

It should be noted that common courses between students' major requirements and the data analytics minor requirements count toward both the major and the minor. The student must complete at least 6 credit hours of coursework that are not counted toward the requirements of their major or any other minor.

Career Options

In today's world, most sectors and industries involve some form of data analysis. Therefore, minor holders would be well-suited to work in various sectors, including but not limited to telecommunication, banking, financial services, environmental, energy, biomedical, police, biology, and physics.

Moreover, minor holders may assume specialized roles such as data scientists, data analysts, and data engineers.

Curriculum and Program

The data analytics minor consists of six courses (18 credits), of which three are mandatory and three are elective, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

I. Mandatory Courses

The three mandatory courses (9 credits) are:

BADM 250/Math 351	Business Statistics/Probability and Statistics	3(3,0)
COSC 480	Machine Learning and Data Mining	3(3,0)
CCEE 567 / BADM 420	Optimization / Quantitative Methods for Business	3(3,0)

II. Elective Courses

Students may choose three elective courses (9 credits) from the following list or any other approved course.

COSC 316	Design and Analysis of Algorithms	3(3,0)
COSC 481	Artificial Intelligence	3(3,0)
COSC 482	Data Science and Web Scraping	3(3,0)
CCEE 516	Advanced Programming and Database Management Systems	3(2,2)
CCEE 568	Big data and analytics/Big data system	3(3,0)
CCEE 561	Computer Vision	3(3,0)
CCEE 566	Natural Language Processing	3(3,0)
CCEE 612	Advanced-Data Mining	3(3,0)
BITM 320	Enterprise Resource Planning	3(3,0)
BITM 415	Business Intelligence	3(3,0)
BMKA 420	Digital and Social Media Marketing	3(3,0)

It should be noted that CBA students (excluding BITM students) can also choose from the following list:

COSC 214	Introduction to Programming	3(3,0)
COSC 231	Database Management Systems	3(3,0)
COSC 333	Web Programming	3(3,0)

III. Courses Description

BADM 250	**Business Statistics**	**3(3,0)**
This course introduces students to the fundamentals of applied multivariate statistics. As such, the course covers factor analysis, multiple regression analysis, discriminant analysis, analysis of variance, and hypothesis testing.		

BADM 420	**Quantitative Methods for Business**	**3(3,0)**
This course introduces students to managerial decision analysis using quantitative tools. The course will introduce students to the use and development of mathematical models to help managers make informed decisions. The focus is on the applied aspects of statistics and math. As such, the course will cover the basics of probabilistic and statistical techniques, decision analysis, linear programming, optimization, forecasting, and waiting-line theory. **Prerequisite**: BADM 250, MATH 351, or Senior Standing.		

BITM 350	**Fundamentals of Data Analytics**	**3(3,0)**
This course introduces students to statistical techniques for analyzing large datasets. The course covers the theory and application of both parametric and nonparametric methods. Students will learn to visualize the data using univariate and bivariate plots and to investigate whether correlations exist in a multidimensional space using factor		

and cluster analyses. They will also build and test predictive models, including linear, logistic, and time-series models. **Prerequisite**: BADM 250 or MATH 351.

BITM 415	**Business Intelligence**	**3(3,0)**
This course introduces business intelligence as computerized support for managerial decision-making. It concentrates on the theoretical and conceptual foundations of business intelligence and commercial tools and techniques available for effective decision support. It focuses on extracting business intelligence from data sets for various applications, such as reporting and visual analytics across domains like web and business analytics, to support decision-making. Provides hands-on experience with various business intelligence software for reporting and building visualizations and dashboards**. Prerequisite**: Senior Standing & BADM 350		

BMKA 420	**Digital and Social Media Marketing**	**3(3,0)**
Digital marketing has evolved from a peripheral element of organizational marketing to the hub of customer-centric communications in an increasingly multi-channel environment. This course covers digital marketing topics like social media, email and mobile marketing, search engine optimization, paid search, and content marketing. It explains the principles of digital marketing and the major factors involved in implementing, measuring, and evaluating successful campaigns that utilize digital marketing channels. **Prerequisite**: Senior Standing		

CCEE 516	**Advanced Programming and Database Management Systems**	**3(3,0)**
This course will introduce students to database management, querying, and data warehousing. Students will also learn advanced programming tools, including Bigtable, NoSQL, R, Python, Scala, MapReduce, and Elasticsearch, and apply them to address big data challenges. **Prerequisite**: CCEE 315 or equivalent.		

CCEE 561	**Computer Vision**	**3(3,0)**
This course introduces the principles, models, and applications of computer vision. The course will cover image structure, projection, stereo vision, and the interpretation of visual motion. Case studies of industrial (robotic) computer vision applications, including visual navigation for autonomous robots, robot hand-eye coordination, and novel man-machine interfaces. **Prerequisite**: CCEE 214 or COSC 214.		

CCEE 566	**Natural Language Processing**	**3(3,0)**
This course introduces the student to natural language processing (NLP). The student is first introduced to word and sentence tokenization. The student then uses the learned skills to implement systems for text classification and sentiment analysis, spelling correction, information extraction, parsing, meaning extraction, and question answering, Machine learning algorithms, as well as algorithms like n-gram language modeling, naive Bayes and maxent classifiers, sequence models like Hidden Markov		

Models, probabilistic dependency and constituent parsing, and vector-space models of meaning will be introduced as needed for the above NLP applications. **Prerequisite**: CCEE 214 or COSC 214.

CCEE 567	**Optimization**	**3(3,0)**
This course introduces students to the theory, algorithms, and applications of optimization. The optimization methodologies include linear programming, network optimization, integer programming, and decision trees. It includes a team project in which students select and solve a problem in practice. **Equivalent to** BADM 420 and MECH 571. **Prerequisite**: MATH 215 or equivalent.		

CCEE 568	**Big Data and Analytics/Big Data System**	**3(3,0)**
This course enables students to understand why the Big Data Era has emerged. Students will become conversant in the terminology and core concepts of big data problems, applications, and systems. Students will learn how to make Big Data useful in their business or career. Students will be introduced to one of the most common frameworks, Hadoop, which has made big data analysis easier and more accessible -- increasing the potential for data to transform our world. **Prerequisite**: CCEE 564 or equivalent.		

CCEE 612	**Advanced-Data Mining**	**3(3,0)**
In this course, advanced data mining topics will be covered, namely, classification (linear discriminant analysis, quadratic discriminant analysis, neural networks), combining multiple learners (bagging, boosting, cascading, stacking), dimensionality reduction (principal component analysis, linear discriminant analysis, subset selection), deep learning, anomaly detection, and reinforcement learning. **Prerequisite**: CCEE 564 or equivalent.		

COSC 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

COSC 316	**Design and Analysis of Algorithms**	**3(3,0)**
This course consolidates algorithm design and programming techniques. It provides an extended study of object-oriented programming properties, data structures, and data abstraction, as well as an introduction to complexity considerations. **Prerequisite***:* COSC 215.		

COSC 333	**Web Programming**	**3(3,0)**
This course teaches students how to develop and implement web-based programs, with an emphasis on front-end programming. It introduces students to web development, client-side languages, and the styles needed to build functional,		

responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, and responsive design. **Prerequisite***:* COSC 214. Equivalent to CCEE 411.

COSC 480	**Machine Learning and Data Mining**	**3(3,0)**
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process, applying appropriate approaches to solve practical problems, and developing intelligent software applications. The course covers several topics about classification, prediction, and clustering. **Prerequisites:** COSC 214 and MATH 351 or BADM 250. Equivalent to CCEE 564.		

COSC 481	**Artificial Intelligence**	**3(3,0)**
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include Intelligent agents, informed and uninformed search strategies, and adversarial search. The Python language libraries will also be introduced. **Prerequisite**: COSC 214. Equivalent to CCEE 562.		

COSC 482	**Data Science and Web Scraping**	**3(3,0)**
Data is becoming the fuel of the 21st century, and acquiring data processing and analysis skills is becoming necessary. This course introduces data science processes, with a focus on web scraping as an application. The course will combine different domains, i.e., web programming, system programming, and machine learning. In particular, the course focuses on analyzing webpage HTML using Python, extracting available information, and generating dashboards. **Co-requisite:** COSC 333 or CCEE 411 and MATH 351 or BADM 250.		

MATH 351	**Probability and Statistics**	**3(3,0)**
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		